

GCSE Mathematics

Predicted Paper 2 — Higher Tier

Set C

Pearson Edexcel style (1MA1 / 2H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

- Calculator
- Ruler graduated in centimetres and millimetres
- Protractor
- Pair of compasses
- Pen, HB pencil, eraser

Instructions

- Answer **all** 22 questions in the spaces provided.
 - You must show all your working.
- Diagrams are NOT accurately drawn unless stated otherwise.
 - Calculators may be used.
- If your calculator does not have a π button, take π to be 3.142.

Advice

- Read each question carefully before you start.
- The marks for each question are shown in brackets — use them as a time guide.
- Try to answer every question. Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

1. The population of a town decreased from 8400 to 7350 over five years. Work out the percentage decrease in population.

(3 marks)

2. Work out $2\frac{3}{4} \times 1\frac{5}{8}$. Give your answer as a mixed number in its simplest form.

(3 marks)

3. The mass of one atom of carbon is 1.99×10^{-23} grams. Work out the total mass of 4×10^{22} atoms of carbon. Give your answer in grams in standard form.

(3 marks)

4. A force of 240 newtons acts on an area of 0.3 m^2 .

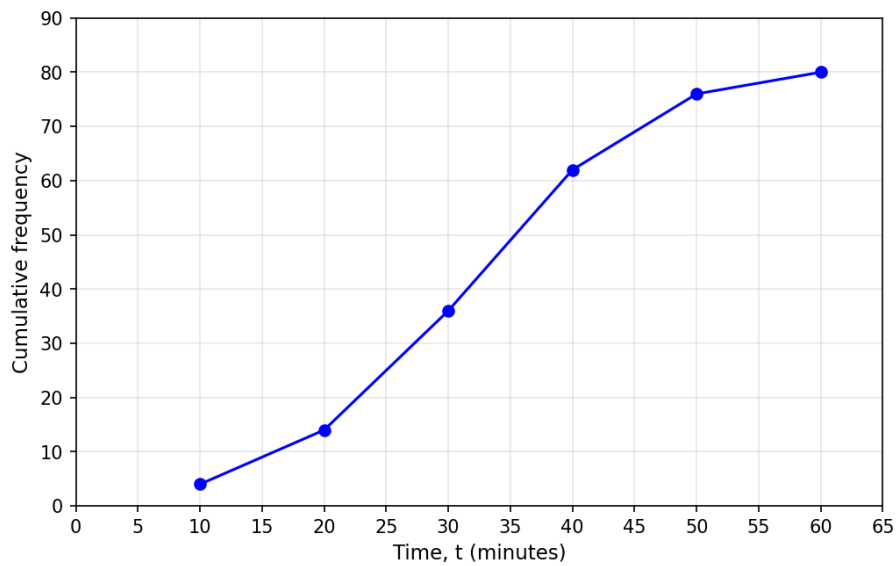
(a) Work out the pressure exerted, in newtons per square metre (N/m^2).

(2 marks)

(b) The same force now acts on a smaller area of 0.15 m^2 . What happens to the pressure? Justify your answer.

(2 marks)

5. The cumulative frequency graph shows information about the time, in minutes, taken by 80 people to complete a task.



(a) Use the graph to find an estimate for the median time.

(1 marks)

(b) Use the graph to find an estimate for the interquartile range.

(2 marks)

(c) Estimate the number of people who took more than 45 minutes to complete the task.

(2 marks)

6. Solve the simultaneous equations:

$$y = x + 3$$

$$y = x^2 + x - 1$$

(4 marks)

7. Simplify $(\sqrt{48} + \sqrt{12}) / \sqrt{3}$. Give your answer in the form a , where a is an integer.

(3 marks)

8. A boat sails 12 km on a bearing of 075° from point P to point Q. It then sails 18 km on a bearing of 165° from Q to point R.

(a) Show that angle $PQR = 90^\circ$.

(2 marks)

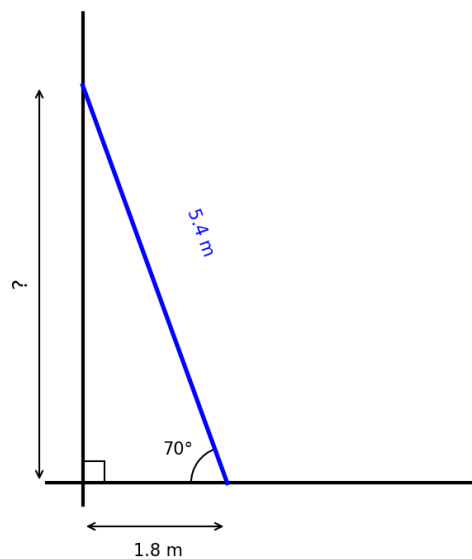
(b) Work out the distance PR. Give your answer to 3 significant figures.

(2 marks)

(c) Work out the bearing of R from P. Give your answer to the nearest degree.

(3 marks)

9. The diagram shows a ladder leaning against a vertical wall. The ladder is 5.4 m long and makes an angle of 70° with the ground. The foot of the ladder is 1.8 m from the wall.



(a) Use trigonometry to work out how high up the wall the ladder reaches. Give your answer to 3 significant figures.

(2 marks)

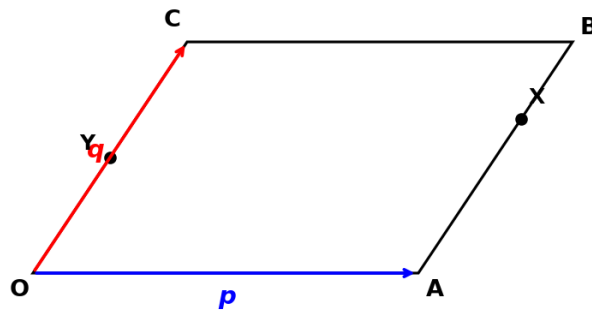
(b) The safety guidance states that the angle between a ladder and the ground should be between 70° and 75° . Use the dimensions given to verify whether this ladder is within safety guidance.

(2 marks)

10. Expand and simplify $(x - 2)(x + 3)(x - 5)$.

(3 marks)

11. OABC is a parallelogram, where $\mathbf{OA} = \mathbf{p}$ and $\mathbf{OC} = \mathbf{q}$. X is the point on AB such that $AX : XB = 2 : 1$. Y is the midpoint of OC.



(a) Express the vector $\mathbf{XY}$ in terms of $\mathbf{p}$ and $\mathbf{q}$. Give your answer in its simplest form.

(4 marks)

(b) Z is a point on the line OB such that $\mathbf{OZ} = k\mathbf{OB}$. Given that X, Y and Z are collinear, find the value of k.

(3 marks)

12. Alice rolls a fair six-sided dice. Ben spins a fair five-sided spinner numbered 1 to 5. The total score is the sum of the dice and the spinner.

(a) Work out the probability that the total score is 7.

(3 marks)

(b) Given that the total score is at least 9, work out the probability that Alice rolled a 6.

(3 marks)

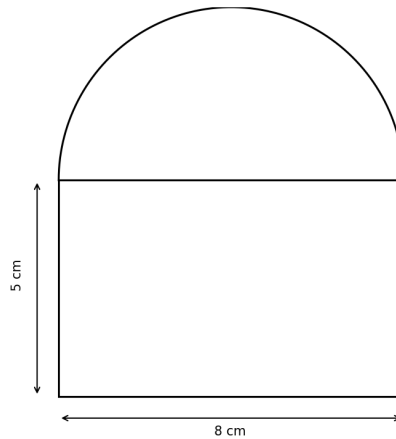
13. Line A has equation $y = 2x + 5$. Line B is perpendicular to line A and passes through the point (4, 1). Find the equation of line B. Give your answer in the form $y = mx + c$.

(4 marks)

14. Triangle PQR has $PQ = 9.6$ cm, $PR = 11.2$ cm, and angle $QPR = 64^\circ$. Work out the area of triangle PQR. Give your answer to 3 significant figures.

(3 marks)

15. The diagram shows a shape made from a rectangle of width 8 cm and height 5 cm, with a semicircle on top (diameter equal to the rectangle's width).



(a) Work out the total area of the shape. Give your answer to 3 significant figures.

(3 marks)

(b) Work out the perimeter of the shape. Give your answer to 3 significant figures.

(3 marks)

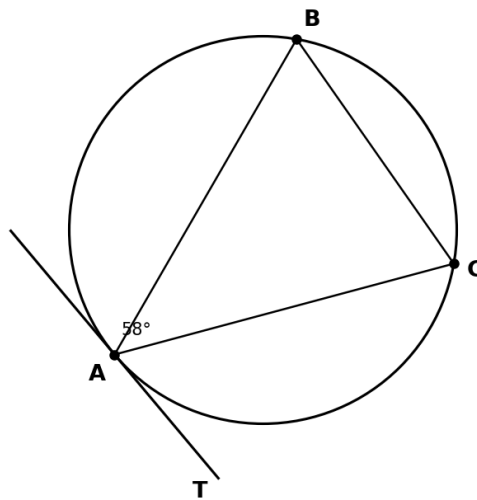
16. Make x the subject of the formula $y = (2x + 3) / (x - 1)$.

(4 marks)

17. Solve $5^{2x} \times 5^{x+1} = 5^{10}$. Show your working.

(3 marks)

18. A, B and C are points on the circumference of a circle. AT is a tangent to the circle at A. Angle TAB = 58° .



(a) Work out the size of angle ACB. Give a reason for your answer.

(2 marks)

(b) If angle ABC = 76° , work out the size of angle BAC.

(2 marks)

19. The pressure P of a gas is directly proportional to the temperature T (in Kelvin) and inversely proportional to the volume V . When $T = 300$ and $V = 5$, $P = 60$.

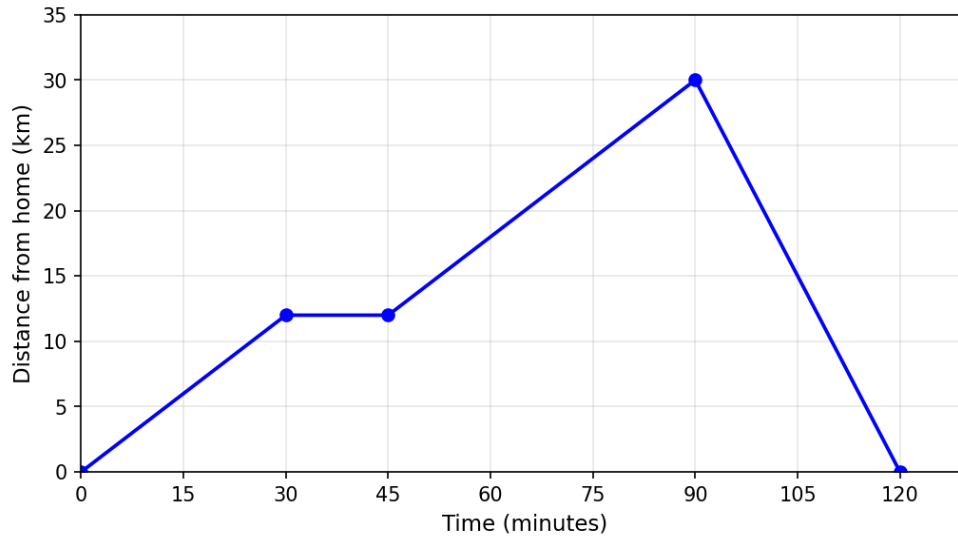
(a) Find a formula for P in terms of T and V .

(3 marks)

(b) Find the value of P when $T = 360$ and $V = 4$.

(2 marks)

20. The travel graph shows information about a cyclist's journey.



(a) Work out the cyclist's speed during the first part of the journey (0 to 30 minutes). Give your answer in km/h.

(2 marks)

(b) The cyclist's average speed for the entire journey was greatest during which part? Justify your answer with calculation.

(3 marks)

21. The graph of $y = f(x)$ where $f(x) = x^2 - 4$ is shown (you may sketch it). On the same axes, sketch the graph of:

(a) $y = f(x) + 3$

(2 marks)

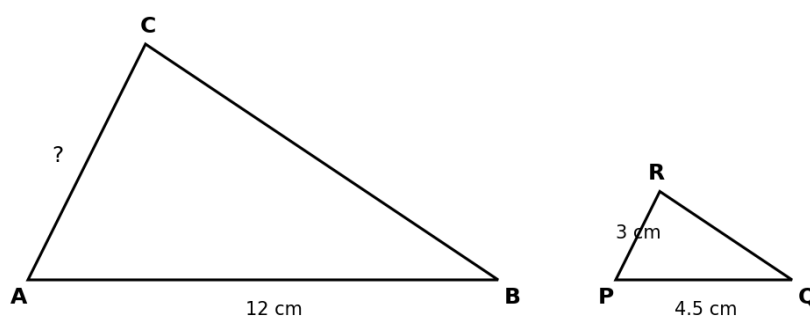
(b) $y = f(x - 2)$

(2 marks)

(c) Write down the coordinates of the turning point of $y = f(x - 2) + 3$.

(1 marks)

22. Triangles ABC and PQR are similar. $AB = 12\text{ cm}$ and $PQ = 4.5\text{ cm}$. The area of triangle PQR is 27 cm^2 .



(a) Work out the scale factor (length) from PQR to ABC.

(1 marks)

(b) Work out the area of triangle ABC.

(3 marks)

TOTAL FOR PAPER: 80 MARKS

END

Mark Scheme — Predicted Paper 2 Set C

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	12.5%	M1: decrease = $8400 - 7350 = 1050$. M1: $1050/8400 \times 100$. A1: 12.5. [3]
2	$4 \frac{15}{32}$	M1: converts to improper fractions: $\frac{11}{4} \times \frac{13}{8}$. M1: multiplies: $(11 \times 13) / (4 \times 8) = 143/32$. A1: $143/32 = 4 \frac{15}{32}$. [3]
3	7.96×10^{-1} grams (= 0.796 g)	M1: $(1.99 \times 4) \times (10^{-23} \times 10^{22}) = 7.96 \times 10^{-1}$. M1: handles powers correctly. A1: 7.96×10^{-1} in correct standard form. [3]
4(a)	800 N/m^2	M1: $240 / 0.3$. A1: 800. [2]
4(b)	Pressure doubles to 1600 N/m^2	M1: $240 / 0.15 = 1600$ (or recognises that halving the area doubles the pressure). A1: clear conclusion that pressure doubles. [2]
5(a)	Median ≈ 30 minutes	B1: reads at cumulative frequency 40 to get median ≈ 31 (accept 28–33). [1]
5(b)	IQR ≈ 20 minutes	M1: reads Q1 at cf = 20 (≈ 23) and Q3 at cf = 60 (≈ 39). A1: IQR $\approx 16-20$. [2]
5(c)	≈ 12 people	M1: at t = 45 reads cf ≈ 68 . M1: $80 - 68 = 12$. A1: 12 (accept 10–14). [2]
6	($x = 2, y = 5$) and ($x = -2, y = 1$)	M1: substitutes $y = x + 3$ into the quadratic. M1: $x + 3 = x^2 + x - 1 \Rightarrow x^2 = 4$. M1: $x = \pm 2$. A1: both pairs (2, 5) and (-2, 1). [4]
7	6	M1: $\sqrt{48} = 4\sqrt{3}$ AND $\sqrt{12} = 2\sqrt{3}$. M1: combined = $6\sqrt{3}$, divided by $\sqrt{3}$. A1: 6. [3]
8(a)	Shown: 90°	M1: bearing 075° to 165° difference is 90° (interior angle at Q). A1: clear explanation that the change in bearing equals the interior angle (with diagram or explanation of supplementary/alternate angles). [2]
8(b)	$PR \approx 21.6 \text{ km}$ (3 s.f.)	M1: Pythagoras: $PR^2 = 12^2 + 18^2 = 468$. A1: $PR = \sqrt{468} = 21.6 \text{ km}$. [2]
8(c)	Bearing $\approx 131^\circ$	M1: angle QPR by trigonometry: $\tan(\text{angle}) = 18/12 \Rightarrow \text{angle} = 56.31^\circ$. M1: bearing of R from P = $075 + 56 = 131^\circ$. A1: 131° (nearest degree). [3]
9(a)	5.07 m	M1: $\sin 70 = h/5.4$, so $h = 5.4 \times \sin 70$. A1: 5.07 (3 s.f.). [2]
9(b)	Within guidance	M1: $\cos(\text{angle}) = 1.8/5.4 = 0.333 \Rightarrow \text{angle} = 70.5^\circ$. A1: 70.5° is within $70-75^\circ$, so within guidance. [2]
10	$x^3 - 4x^2 - 11x + 30$	M1: expands two brackets first: $(x - 2)(x + 3) = x^2 + x - 6$. M1: multiplies by $(x - 5)$. A1: $x^3 - 4x^2 - 11x + 30$. [3]
11(a)	$\mathbf{XY} = -\mathbf{p} - (1/6)\mathbf{q}$	M1: $\mathbf{OX} = \mathbf{p} + (2/3)\mathbf{q}$. M1: $\mathbf{OY} = (1/2)\mathbf{q}$. M1: $\mathbf{XY} = \mathbf{OY} - \mathbf{OX}$. A1: $(1/2)\mathbf{q} - \mathbf{p} - (2/3)\mathbf{q} = -\mathbf{p} - (1/6)\mathbf{q}$. [4]
11(b)	$k = 3/5$	M1: $\mathbf{OB} = \mathbf{p} + \mathbf{q}$, $\mathbf{OZ} = k\mathbf{p} + k\mathbf{q}$. M1: $\mathbf{XZ} = (k - 1)\mathbf{p} + (k - 2/3)\mathbf{q}$. M1: sets $\mathbf{XZ}$ parallel to $\mathbf{XY}$: ratios of coefficients equal. A1: $k = 3/5$. [3]
12(a)	$5/30 = 1/6$	M1: lists outcomes summing to 7: (2,5), (3,4), (4,3), (5,2), (6,1). M1: total outcomes = 30. A1: $5/30 = 1/6$. [3]
12(b)	$1/2$	M1: outcomes with total ≥ 9 : (4,5), (5,4), (5,5), (6,3), (6,4), (6,5) — 6 outcomes. M1: of these, 3 have Alice rolling 6: (6,3), (6,4), (6,5). A1: $P = 3/6 = 1/2$. [3]
13	$y = -(1/2)x + 3$	M1: gradient of B = $-1/2$ (negative reciprocal). M1: substitutes (4, 1): $1 = -1/2 \times 4 + c \Rightarrow c = 3$. A1: $y = -(1/2)x + 3$. [4]
14	48.3 cm^2	M1: area = $(1/2) \times 9.6 \times 11.2 \times \sin 64$. M1: $= (1/2) \times 107.52 \times 0.8988$. A1: 48.3 (3 s.f.). [3]

Q	Answer	Mark scheme
15(a)	65.1 cm^2	M1: rectangle area = $8 \times 5 = 40$. M1: semicircle area = $(1/2)\pi(4)^2 = 8\pi$. A1: $40 + 8\pi = 65.13\dots \approx 65.1$. [3]
15(b)	30.6 cm	M1: perimeter = 2 sides of rectangle + bottom + semicircle arc = $5 + 5 + 8 + \pi(4)$. M1: arc = $4\pi = 12.57$. A1: $18 + 4\pi = 30.57 \approx 30.6$. [3]
16	$x = (y + 3) / (y - 2)$	M1: $y(x - 1) = 2x + 3$. M1: $yx - y = 2x + 3$. M1: $x(y - 2) = y + 3$. A1: $x = (y + 3)/(y - 2)$. [4]
17	$x = 3$	M1: combines exponents: $5^{2x + x + 1} = 5^{10}$. M1: $3x + 1 = 10$. A1: $x = 3$. [3]
18(a)	Angle ACB = 58°	B1: 58° . B1: reason "alternate segment theorem" (angle between tangent and chord equals angle in the alternate segment). [2]
18(b)	Angle BAC = 46°	M1: angles in triangle ABC sum to 180° ; $180 - 76 - 58 = 46$. A1: 46° . [2]
19(a)	$P = T / V$	M1: $P = kT/V$. M1: $60 = k(300)/5 = 60k$, so $k = 1$. A1: $P = T/V$. [3]
19(b)	$P = 90$	M1: $P = 360 / 4$. A1: 90. [2]
20(a)	24 km/h	M1: 12 km in 30 min = 12 km in 0.5 h. A1: $12/0.5 = 24 \text{ km/h}$. [2]
20(b)	Return journey (90–120 min)	M1: return journey: 30 km in 30 min = $30/0.5 = 60 \text{ km/h}$. M1: compares to first part (24 km/h) and middle part $((30-12)/0.75 = 24 \text{ km/h})$. A1: return journey is fastest. [3]
21(a)	Vertical translation up 3: turning point (0, -1)	B1: correct shape moved up 3. B1: TP at (0, -1). [2]
21(b)	Horizontal translation right 2: turning point (2, -4)	B1: correct shape moved right 2. B1: TP at (2, -4). [2]
21(c)	(2, -1)	B1ft: applies both transformations. [1]
22(a)	Scale factor = $12 / 4.5 = 8/3$	B1: $8/3$ (or 2.67). [1]
22(b)	192 cm^2	M1: area scale factor = $(8/3)^2 = 64/9$. M1: area ABC = $27 \times 64/9$. A1: 192. [3]

Worked Solutions — Predicted Paper 2 Set C

Higher Tier · Edexcel 1MA1 style

Q1: Percentage decrease

$$\text{Decrease} = 8400 - 7350 = 1050.$$

$$\text{Percentage decrease} = (1050 / 8400) \times 100 = \mathbf{12.5\%}.$$

Q2: Fractions multiplication

Convert to improper fractions: $2 \frac{3}{4} = \frac{11}{4}$ and $1 \frac{5}{8} = \frac{13}{8}$.

$$\text{Product} = (\frac{11}{4}) \times (\frac{13}{8}) = \frac{143}{32}.$$

Convert back to mixed number: $143 \div 32 = 4$ remainder 15, so = **4 $\frac{15}{32}$** .

Q3: Standard form

$$\text{Total mass} = (1.99 \times 10^{-23}) \times (4 \times 10^{22}).$$

$$= (1.99 \times 4) \times (10^{-23} \times 10^{22})$$

$$= 7.96 \times 10^{-1} \text{ grams } (= 0.796 \text{ g}).$$

Q4: Pressure

$$\text{(a) Pressure} = \text{force} / \text{area} = 240 / 0.3 = \mathbf{800 \text{ N/m}^2}.$$

$$\text{(b) New pressure} = 240 / 0.15 = \mathbf{1600 \text{ N/m}^2}.$$

Since the area was halved, the pressure has **doubled**.

Q5: Cumulative frequency

(a) Median is at cumulative frequency $80/2 = 40$. Reading across: $t \approx \mathbf{31 \text{ minutes}}$.

(b) Q1 at cf = 20: $t \approx 23 \text{ min}$. Q3 at cf = 60: $t \approx 39 \text{ min}$.

$$\text{IQR} \approx 39 - 23 = \mathbf{16 \text{ minutes}}.$$

(c) At $t = 45 \text{ min}$: cf ≈ 68 . People who took longer = $80 - 68 = \mathbf{12 \text{ people}}$.

Q6: Linear-quadratic simultaneous

Substitute $y = x + 3$ into $y = x^2 + x - 1$:

$$x + 3 = x^2 + x - 1 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2.$$

When $x = 2$: $y = 5$. When $x = -2$: $y = 1$.

Solutions: **(2, 5) and (-2, 1)**.

Q7: Surds

$$\sqrt{48} = \sqrt{(16 \times 3)} = 4\sqrt{3}. \quad \sqrt{12} = \sqrt{(4 \times 3)} = 2\sqrt{3}.$$

$$\text{Sum: } 4\sqrt{3} + 2\sqrt{3} = 6\sqrt{3}.$$

$$\text{Divide by } \sqrt{3}: 6\sqrt{3} / \sqrt{3} = \mathbf{6}.$$

Q8: Bearings

(a) At Q, the original bearing is 075° (direction PQ extended). The new bearing of R from Q is 165° .
The angle between the inbound direction (reverse of PQ, which is bearing $075 + 180 = 255^\circ$) and

the outbound direction (165°) is $255 - 165 = 90^\circ$.

So **angle PQR = 90°** . ✓

(b) Triangle PQR is right-angled at Q with legs PQ = 12 km and QR = 18 km.

$$PR^2 = 12^2 + 18^2 = 144 + 324 = 468.$$

$$PR = \sqrt{468} = \mathbf{21.6 \text{ km (3 s.f.)}}.$$

(c) $\tan(\text{angle QPR}) = \text{opposite/adjacent} = 18/12 = 1.5$, so angle QPR = 56.3° .

Bearing of R from P = $075 + 56 = \mathbf{131^\circ}$ (nearest degree).

Q9: Right-angled trig

(a) $\sin 70^\circ = (\text{height up wall}) / 5.4$. Height = $5.4 \times \sin 70^\circ = 5.4 \times 0.9397 = \mathbf{5.07 \text{ m (3 s.f.)}}$.

(b) Check angle: $\cos(\text{angle}) = \text{adjacent/hypotenuse} = 1.8 / 5.4 = 0.3333$.

$$\text{Angle} = \cos^{-1}(0.3333) = \mathbf{70.5^\circ}.$$

Since $70^\circ \leq 70.5^\circ \leq 75^\circ$, the ladder is **within safety guidance**.

Q10: Triple bracket expansion

First expand $(x - 2)(x + 3) = x^2 + 3x - 2x - 6 = x^2 + x - 6$.

Multiply by $(x - 5)$: $(x^2 + x - 6)(x - 5)$.

$$= x^3 - 5x^2 + x^2 - 5x - 6x + 30$$

$$= \mathbf{x^3 - 4x^2 - 11x + 30}.$$

Q11: Vectors

(a) $\mathbf{OX} = \mathbf{OA} + \mathbf{AX} = \mathbf{p} + (2/3)\mathbf{AB}$. In parallelogram OABC, $\mathbf{AB} = \mathbf{OC} = \mathbf{q}$. So $\mathbf{OX} = \mathbf{p} + (2/3)\mathbf{q}$.

$$\mathbf{OY} = (1/2)\mathbf{OC} = (1/2)\mathbf{q}.$$

$$\mathbf{XY} = \mathbf{OY} - \mathbf{OX} = (1/2)\mathbf{q} - \mathbf{p} - (2/3)\mathbf{q} = \mathbf{-p - (1/6)q}.$$

(b) $\mathbf{OB} = \mathbf{p} + \mathbf{q}$, so $\mathbf{OZ} = k(\mathbf{p} + \mathbf{q})$.

$$\mathbf{XZ} = \mathbf{OZ} - \mathbf{OX} = k\mathbf{p} + k\mathbf{q} - \mathbf{p} - (2/3)\mathbf{q} = (k - 1)\mathbf{p} + (k - 2/3)\mathbf{q}.$$

For X, Y, Z collinear: $\mathbf{XZ} = \lambda\mathbf{XY} = \lambda(-\mathbf{p} - (1/6)\mathbf{q})$.

Equating: $k - 1 = -\lambda$ and $k - 2/3 = -\lambda/6$.

From the first: $\lambda = 1 - k$. Substitute: $k - 2/3 = -(1 - k)/6$.

$$6(k - 2/3) = -(1 - k) \Rightarrow 6k - 4 = k - 1 \Rightarrow 5k = 3 \Rightarrow \mathbf{k = 3/5}.$$

Q12: Probability (independent)

(a) Total outcomes = $6 \times 5 = 30$. Outcomes summing to 7: (2,5), (3,4), (4,3), (5,2), (6,1) — 5 outcomes.

$$P(\text{sum} = 7) = 5/30 = \mathbf{1/6}.$$

(b) Outcomes with sum ≥ 9 : (4,5), (5,4), (5,5), (6,3), (6,4), (6,5) — 6 outcomes.

Of these, Alice rolled 6 in: (6,3), (6,4), (6,5) — 3 outcomes.

$$P(\text{Alice rolled 6} \mid \text{sum} \geq 9) = 3/6 = \mathbf{1/2}.$$

Q13: Perpendicular line

Line A has gradient 2. The perpendicular gradient is $-1/2$.

Line B: $y = -(1/2)x + c$. Substitute (4, 1): $1 = -2 + c \Rightarrow c = 3$.

$y = -(1/2)x + 3$ (or equivalently $y = -x/2 + 3$).

Q14: Area using sine

Area = $(1/2) \times PQ \times PR \times \sin(\text{angle QPR})$.

= $(1/2) \times 9.6 \times 11.2 \times \sin 64^\circ$

= $(1/2) \times 107.52 \times 0.8988$

= **48.3 cm² (3 s.f.)**.

Q15: Compound shape

(a) Rectangle area = $8 \times 5 = 40 \text{ cm}^2$.

Semicircle (radius 4): area = $(1/2)\pi(4)^2 = 8\pi = 25.13 \text{ cm}^2$.

Total = $40 + 25.13 = \mathbf{65.1 \text{ cm}^2 \text{ (3 s.f.)}}$.

(b) Perimeter: two vertical sides ($5 + 5 = 10$) + bottom (8) + semicircle arc.

Semicircle arc = $\pi \times r = 4\pi = 12.57 \text{ cm}$.

Total = $10 + 8 + 12.57 = \mathbf{30.6 \text{ cm (3 s.f.)}}$.

Q16: Make x the subject

$y(x - 1) = 2x + 3$.

$yx - y = 2x + 3$.

Move x terms to one side: $yx - 2x = y + 3$.

Factor: $x(y - 2) = y + 3$.

$x = \mathbf{(y + 3) / (y - 2)}$.

Q17: Exponential equation

Use exponent rule: $5^{2x} \times 5^{x+1} = 5^{2x+x+1} = 5^{3x+1}$.

So $3x + 1 = 10 \Rightarrow 3x = 9 \Rightarrow \mathbf{x = 3}$.

Q18: Alternate segment theorem

(a) By the **alternate segment theorem**, the angle between the tangent AT and the chord AB equals the angle in the alternate segment, which is angle ACB.

So **angle ACB = 58°**.

(b) Angles in triangle ABC sum to 180°.

Angle BAC = $180 - 76 - 58 = \mathbf{46^\circ}$.

Q19: Compound proportion

(a) $P = kT/V$. Substitute $T = 300$, $V = 5$, $P = 60$: $60 = 300k/5 = 60k \Rightarrow k = 1$.

$P = T/V$.

(b) $P = 360/4 = \mathbf{90}$.

Q20: Distance-time graph

(a) 12 km in 30 minutes = 12 km in 0.5 hours. Speed = $12/0.5 = \mathbf{24 \text{ km/h}}$.

(b) Compare speeds for each leg:

Leg 1 (0–30 min): $12 \text{ km} / 0.5 \text{ h} = 24 \text{ km/h}$.

Leg 2 (30–45 min): stationary, 0 km/h .

Leg 3 (45–90 min): $18 \text{ km} / 0.75 \text{ h} = 24 \text{ km/h}$.

Leg 4 (90–120 min): $30 \text{ km} / 0.5 \text{ h} = 60 \text{ km/h}$.

Fastest is the **return journey (90–120 minutes) at 60 km/h**.

Q21: Graph transformations

$f(x) = x^2 - 4$, turning point at $(0, -4)$.

(a) $y = f(x) + 3$: translates up by 3 units. Turning point = **$(0, -1)$** .

(b) $y = f(x - 2)$: translates right by 2 units. Turning point = **$(2, -4)$** .

(c) $y = f(x - 2) + 3$: right 2, up 3. Turning point = **$(2, -1)$** .

Q22: Similar triangles

(a) Length scale factor $PQR \rightarrow ABC = 12 / 4.5 = \mathbf{8/3}$ (or 2.67).

(b) Area scale factor = $(\text{length scale factor})^2 = (8/3)^2 = 64/9$.

Area of ABC = area of PQR $\times 64/9 = 27 \times 64/9 = \mathbf{192 \text{ cm}^2}$.